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Program:

```
CHANNEL_FILE = "channel.txt"

def text_to_bin(t): return "".join(format(ord(c), "08b") for c in t)
def bin_to_text(b): return "".join(chr(int(b[i:i+8], 2)) for i in range(0,
len(b), 8))

def encode_nibble(d):
    d1, d2, d3, d4 = (int(x) for x in d)
    p1, p2, p4 = d1^d2^d4, d1^d3^d4, d2^d3^d4
    return f"{p1}{p2}{d1}{p4}{d2}{d3}{d4}"

def sender(text):
    bits = text_to_bin(text)
    bits += "0" * (-len(bits) % 4)
    encoded = "".join(encode_nibble(bits[i:i+4]) for i in range(0,
len(bits), 4))
    open(CHANNEL_FILE, "w").write(encoded)
    print("Sent:", encoded)

def check_block(block):
    b = [int(x) for x in block]
    c1 = b[0]^b[2]^b[4]^b[6]
    c2 = b[1]^b[2]^b[5]^b[6]
    c4 = b[3]^b[4]^b[5]^b[6]
    err = c1 + 2*c2 + 4*c4
    if err: b[err-1] ^= 1
    return "".join(map(str, b)), err

def receiver():
    stream = open(CHANNEL_FILE).read().strip()
```

```

data = ""
for i in range(0, len(stream), 7):
    block, err = check_block(stream[i:i+7])
    print(f"Block {i//7+1}: {'error at bit '+str(err) if err else 'ok'}")
    data += block[2] + block[4] + block[5] + block[6]
text = bin_to_text(data)
print("Received:", text)
return text

if __name__ == "__main__":
    msg = input("Enter text to send: ")
    sender(msg)

    choice = input("Inject a bit error before receiving? (y/n): ")
    choice = choice.strip().lower()
    if choice == "y":
        bits_len = len(open(CHANNEL_FILE).read().strip())
        pos = int(input(f"Enter bit index to flip (0 to {bits_len-1}): "))
        s = list(open(CHANNEL_FILE).read().strip())
        s[pos] = "1" if s[pos] == "0" else "0"
        open(CHANNEL_FILE, "w").write("".join(s))
        print(f"Flipped bit at index {pos}")

    out = receiver()
    print("Correct:", out == msg)

```

Output:

Hamming code detection

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main.py

Visualize

Run

```
1 print received, ready
2 return text
3
33
34
35 if __name__ == "__main__":
36     msg = input("Enter text to send: ")
37     channel = sender(msg)
38
39     choice = input("Inject a bit error before receiving? (y/n): ")
40     choice = choice.strip().lower()
41     if choice == "y":
42         pos = int(input(f"Enter bit index to flip (0 to {len(channel)-1}): "))
43         s = list(channel)
44         s[pos] = "1" if s[pos] == "0" else "0"
45         channel = "".join(s)
46         print(f"Flipped bit at index {pos}")
47
48     out = receiver(channel)
49     print("Correct:", out == msg)
```

Output

Enter text to send: Hi!
Sent: 10011001110000110011000110010101010101001
Inject a bit error before receiving? (y/n): n
Block 1: ok
Block 2: ok
Block 3: ok
Block 4: ok
Block 5: ok
Block 6: ok
Received: Hi!
Correct: True

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ENG

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24-09-2026